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Problem 1a

```
% Define the coefficient matrix
A = [1 3 -3 7; 0 1 -4 5];

% Define the vectors from the parametric vector form
u = [-9; 4; 1; 0];
v = [8; -5; 0; 1];

% Set the parameters (note that these can be any real numbers!)
s = 10;
t = -1/2;

% Calculate the solution vector from parametric vector form
x = s*u + t*v;

% Verify that A*x gives the zero vector. Note that we
% want the zero vector to appear because we solved the
% homogeneous system Ax=0
z = A*x

z =

    0
    0
```

Problem 1b

```
A = [1 5 2 -6 9 0;
     0 0 1 -7 4 -8;
     0 0 0 0 0 1;
     0 0 0 0 0 0];

u = [-5; 1; 0; 0; 0; 0];
v = [-8; 0; 7; 1; 0; 0];
w = [-1; 0; -4; 0; 1; 0];

s = -5;
t = 11;
q = 1;
```

```
x = s*u + t*v + q*w;
```

```
z = A*x
```

```
z =
```

```
0  
0  
0  
0
```

Problem 3a

```
A = [1 3 1;  
     -4 -9 2;  
     0 -3 -6];
```

```
u = [-2; 1; 0];  
v = [5; -2; 1];
```

```
s = 1;  
t = -5;
```

```
x = s*u + t*v;
```

```
% Note that here, we get the vector [1, -1, -3]  
% because that is the b vector of the  
% nonhomogeneous system!  
z = A*x
```

```
z =
```

```
1  
-1  
-3
```

Problem 3b

```
A = [1 0 -2 -5;  
     -1 1 3 7;  
     3 -2 2 1];
```

```
u = [1; 2; 0; 0];  
v = [1; 0; -2; 1];
```

```
s = 5;
```

```
% Note that u doesn't get multiplied by anything here  
x = u + s*v;
```

$$z = A \cdot x$$

$$z =$$

$$\begin{array}{c} 1 \\ 1 \\ -1 \end{array}$$

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